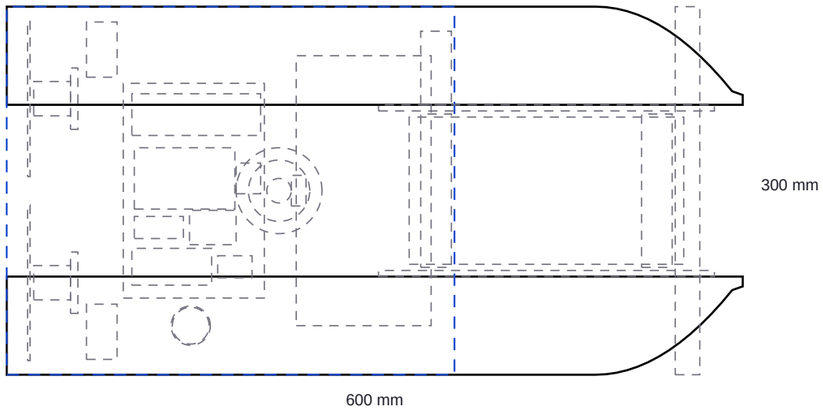
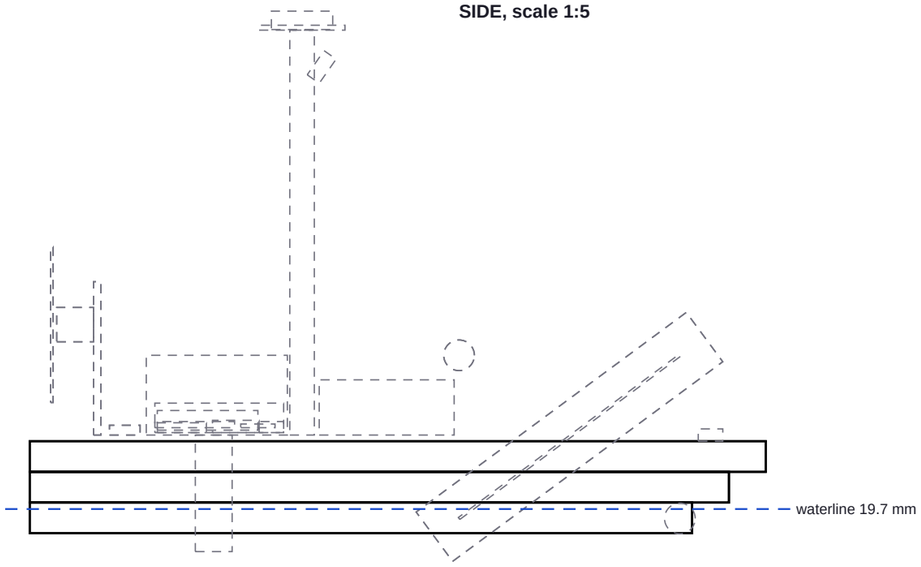
VARUNA Conveyor Cat

600 x 300 mm thermocol boat | Advanced

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 426 mm to GPS top
Weight	1563 g empty, 1781 g with 218 g of plastic
Draft	19.7 mm empty, 22.5 mm full
Reserve buoyancy	4.5 kg before the deck goes under
Stability (GM)	534 mm: very stable
Trim	bow 13.1 mm higher than stern empty; slide the battery to level
Collection	1.09 L deck bin
Camera	360 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	1 sheet 1000 x 500 x 25 mm

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

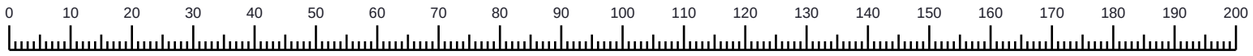
cut line

cut out (window / pocket)

fold / glue line

placement guide, do not cut

skewer hole: lines up the layers



ruler in mm: the 0 to 200 marks must be exactly 200 mm apart

100 mm x 100 mm

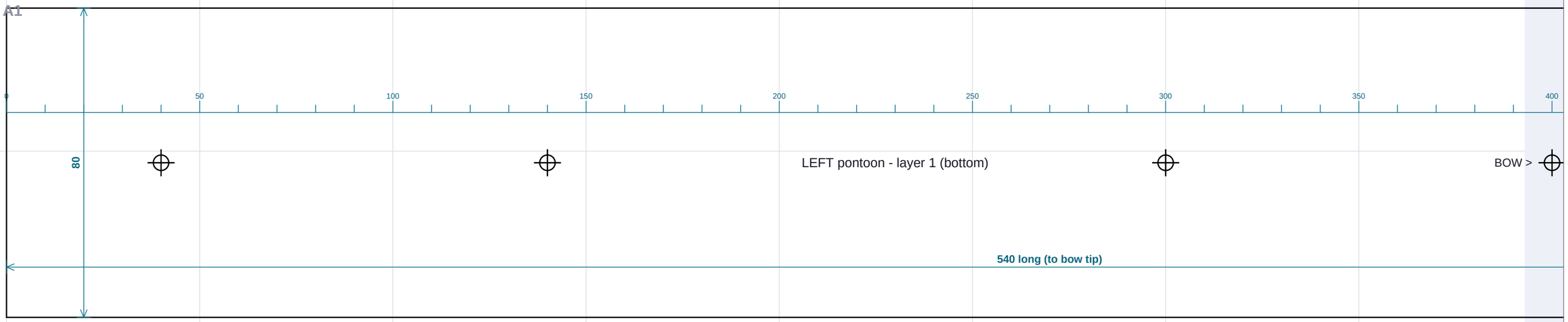
How to print on A3 so every part comes out real size

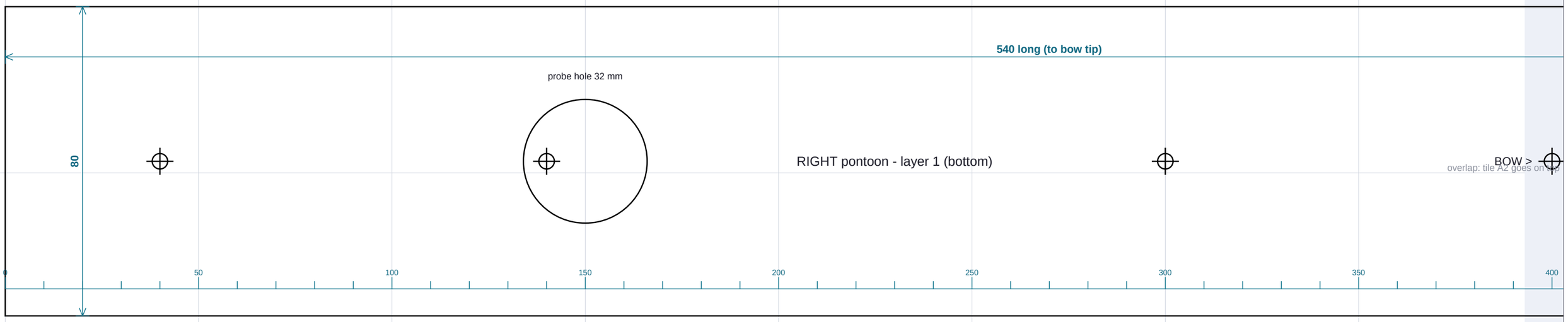
1. Paper: A3, landscape. Printer: 'A3' selected as the paper size.
2. Scaling: 'Actual size' or '100%'. Turn OFF 'Fit to page', 'Shrink oversized pages' and 'Fit to printable area'. On a phone, choose 'Original size' / 'Do not scale'.
3. Print this page only. Measure the square with a steel ruler: both sides must be 100 mm, and the ruler marks must match your ruler. 1 mm off is fine; more is not.
4. If it is wrong, fix the printer setting and print this page again. Do not go on.
5. When it is right, print all the other pages with the SAME settings.
6. Big parts are split across pages (tiles A1, A2, B1...). Cut each tile on its border, lay the next tile over the grey overlap strip, line up the grey 50 mm grid and the measurement ruler, and tape it. Check the long measurement with a tape before cutting.
7. Glue-stick the paper on the thermocol, cut along the solid black line, cut out the red parts, and push a skewer through every + mark.

Line types:

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers
- measurement (mm), do not cut

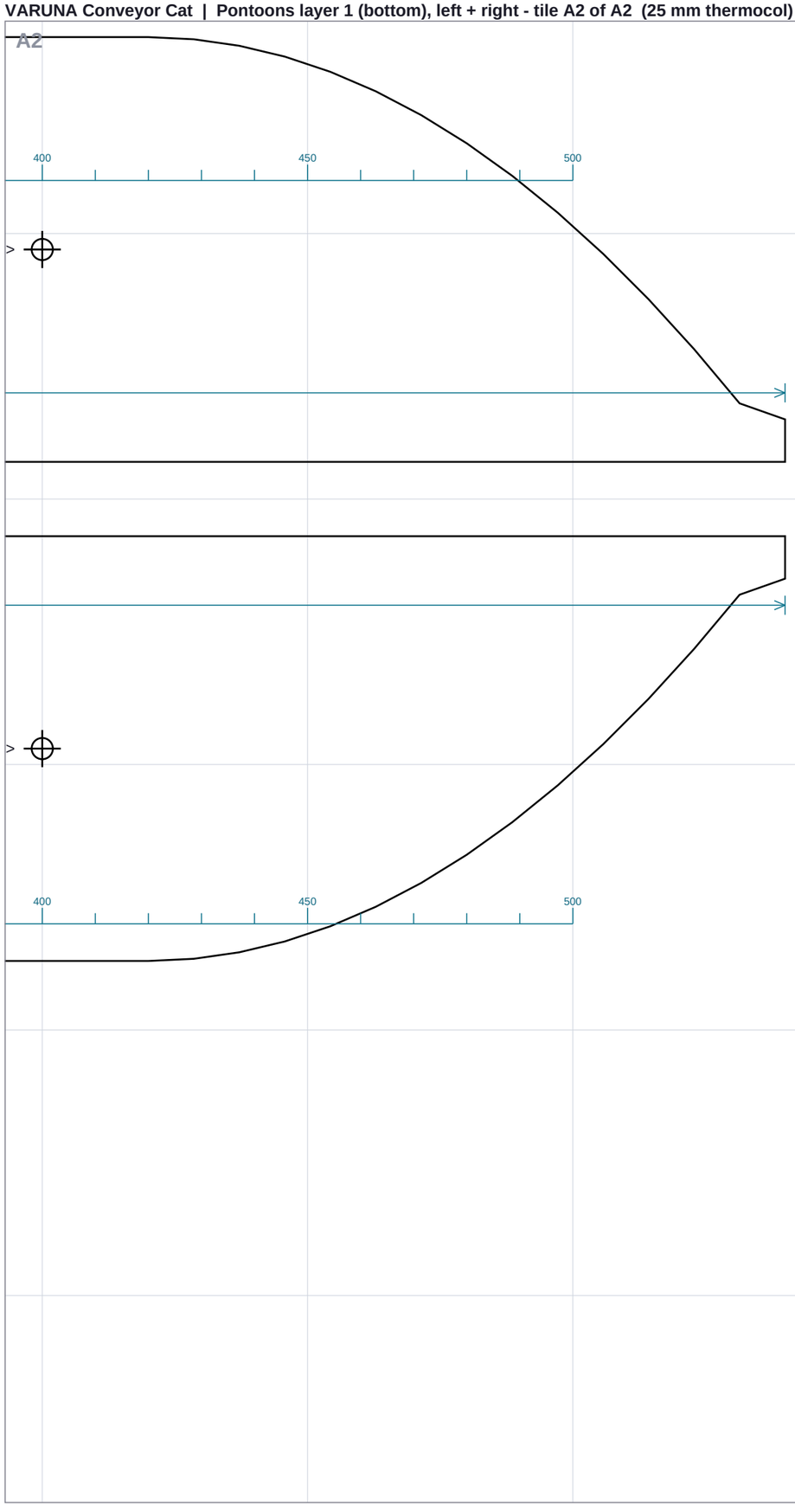
A1



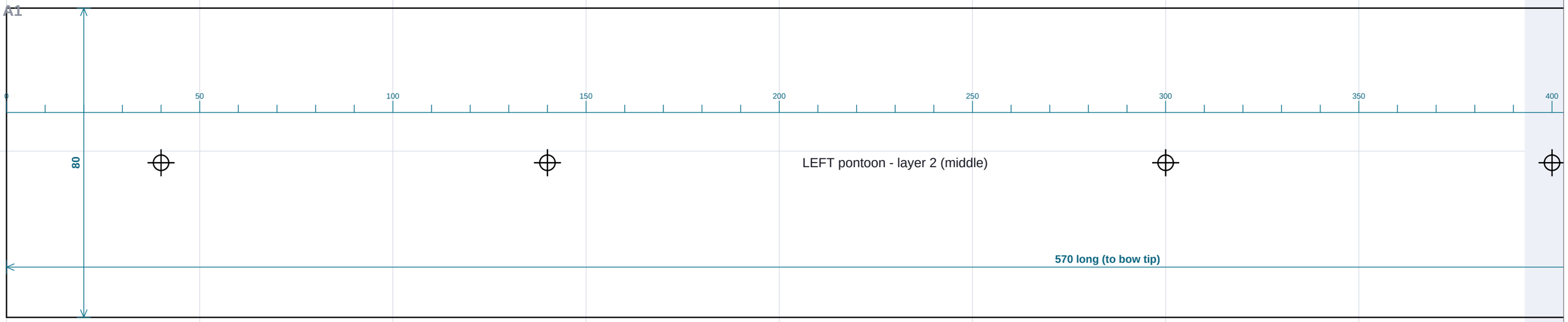


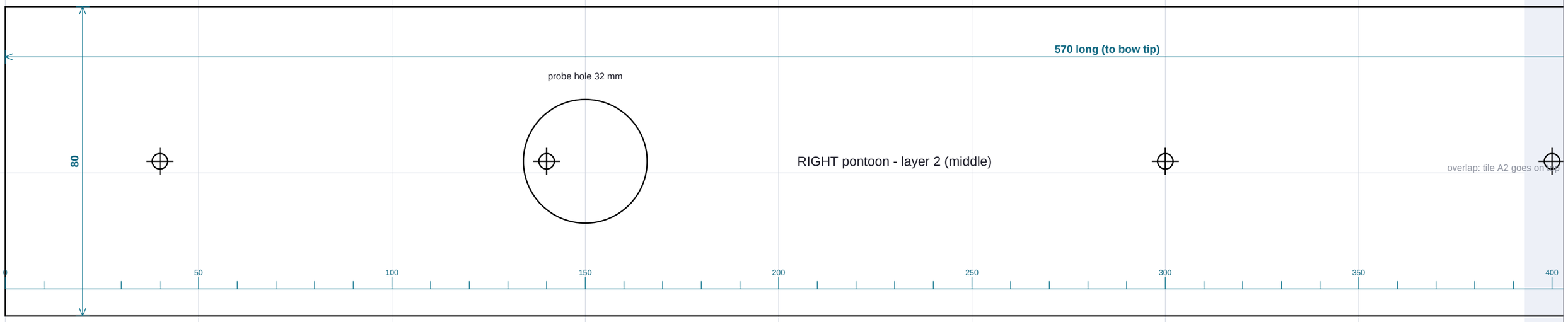
overlap: tile A2 goes on top

<- must measure 50 mm



A1


0
50
100
150
200
250
300
350
400
80
LEFT pontoon - layer 2 (middle)
570 long (to bow tip)


570 long (to bow tip)
probe hole 32 mm
80
RIGHT pontoon - layer 2 (middle)
overlap: tile A2 goes on top
0
50
100
150
200
250
300
350
400

A2

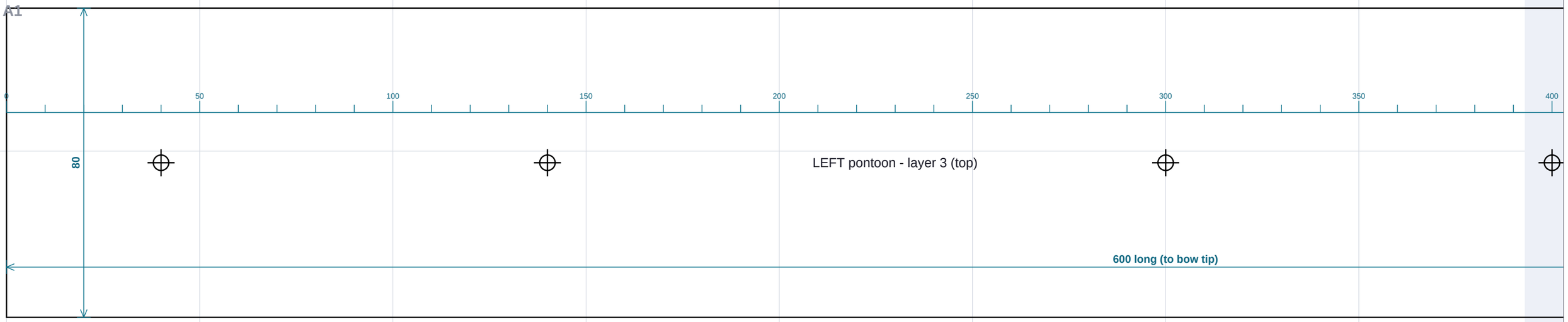
400 450 500

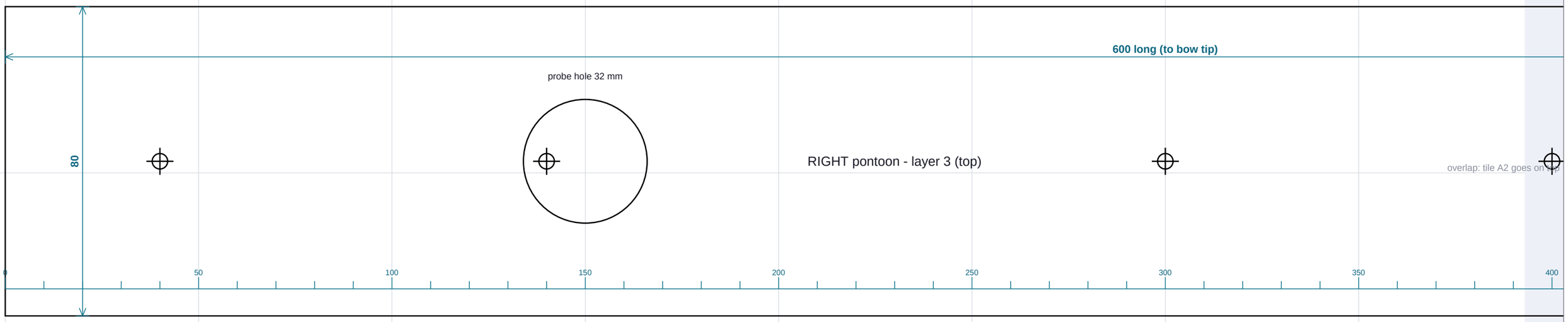
BOW >

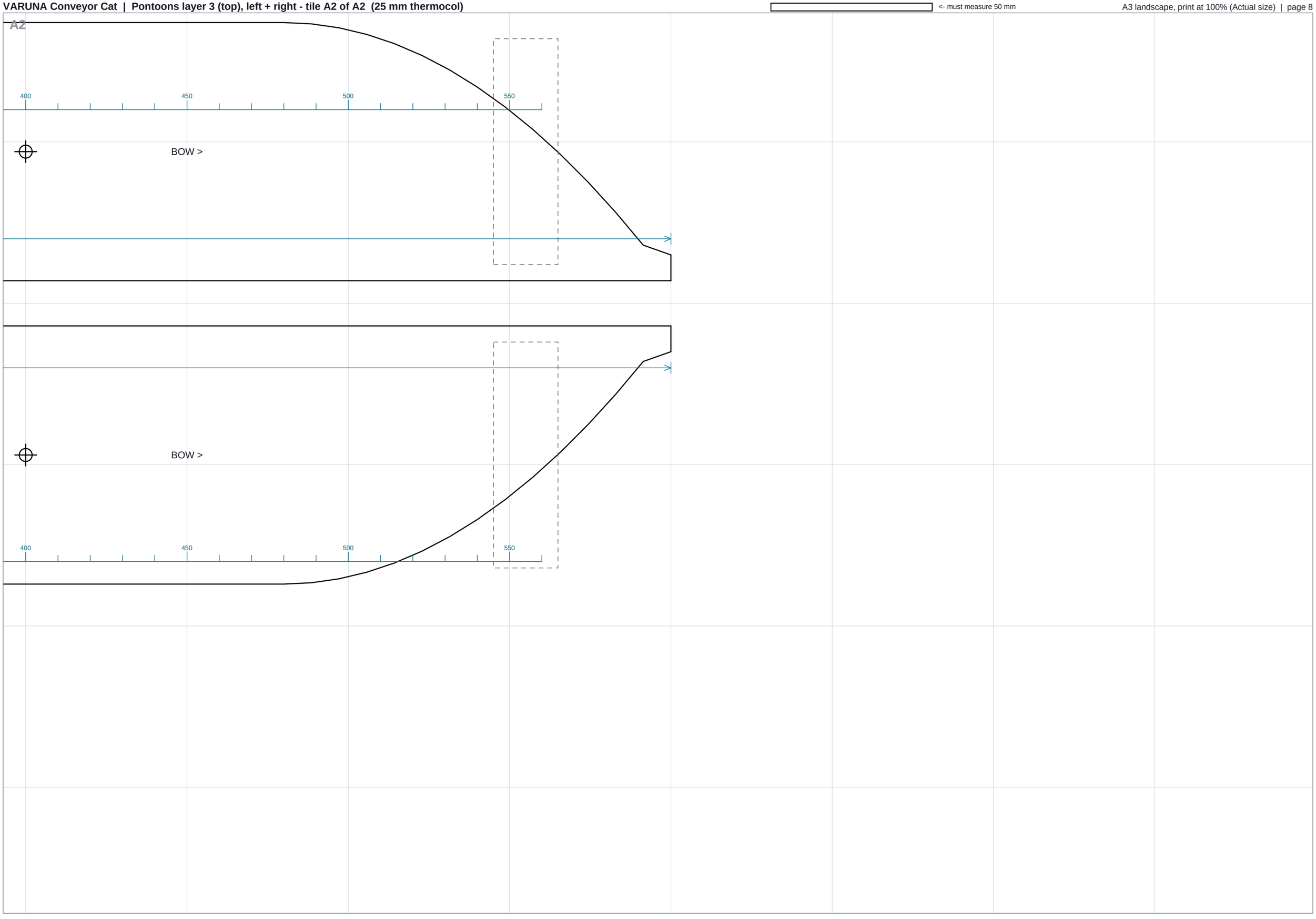
BOW >

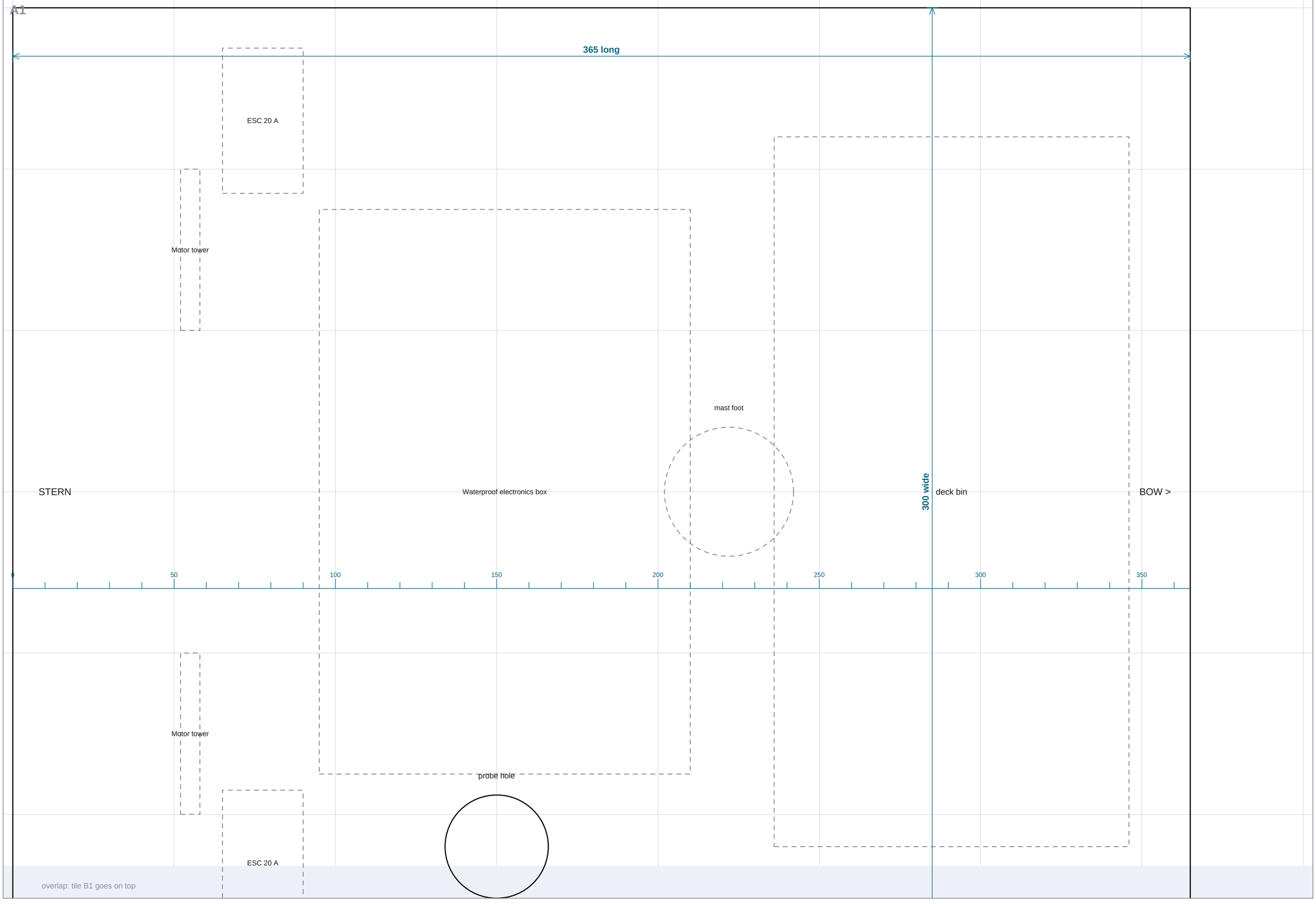
400 450 500

A1


0
50
100
150
200
250
300
350
400
80
LEFT pontoon - layer 3 (top)
600 long (to bow tip)


600 long (to bow tip)
probe hole 32 mm
80
RIGHT pontoon - layer 3 (top)
overlap: tile A2 goes on top
0
50
100
150
200
250
300
350
400


A2
400
450
500
550
BOW >
BOW >
400
450
500
550


A1
365 long
ESC 20 A
Motor tower
mast foot
STERN
Waterproof electronics box
300 wide
deck bin
BOW >
0
50
100
150
200
250
300
350
Motor tower
probe hole
ESC 20 A
overlap: tile B1 goes on top

<- must measure 50 mm

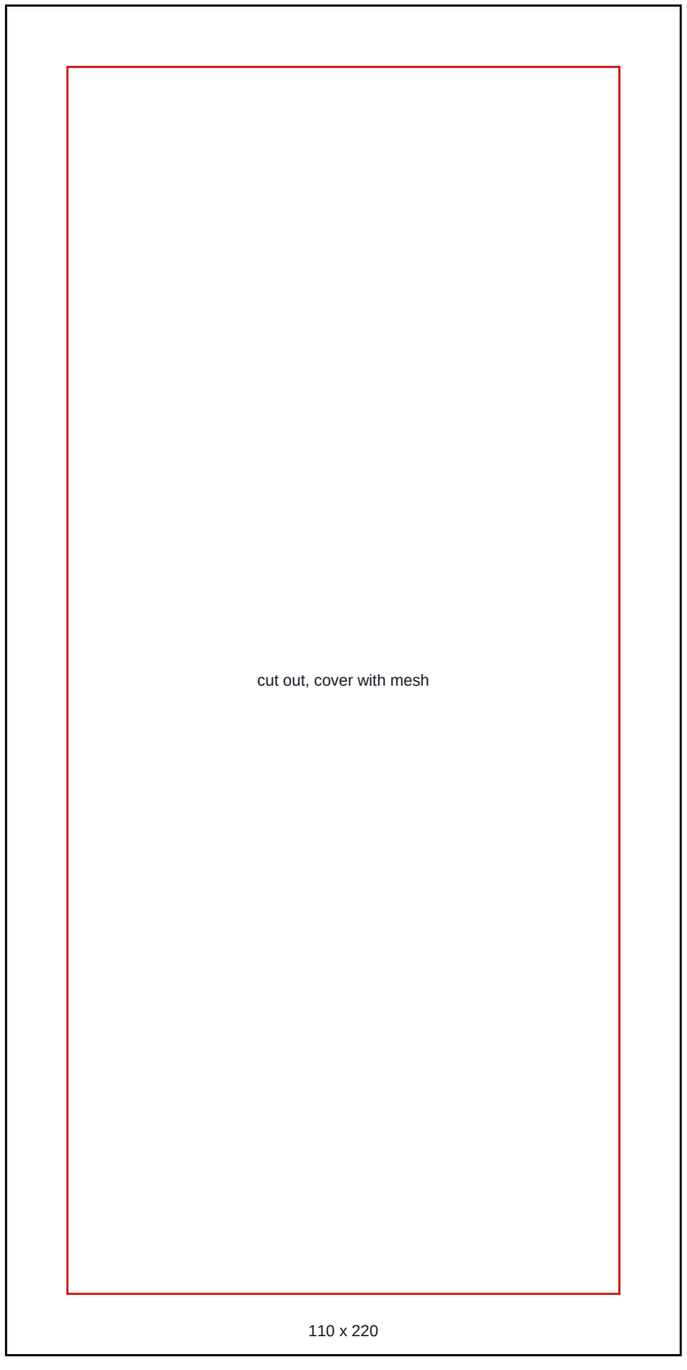
B1

VARUNA Conveyor Cat | small parts 1 of 4

<- must measure 50 mm

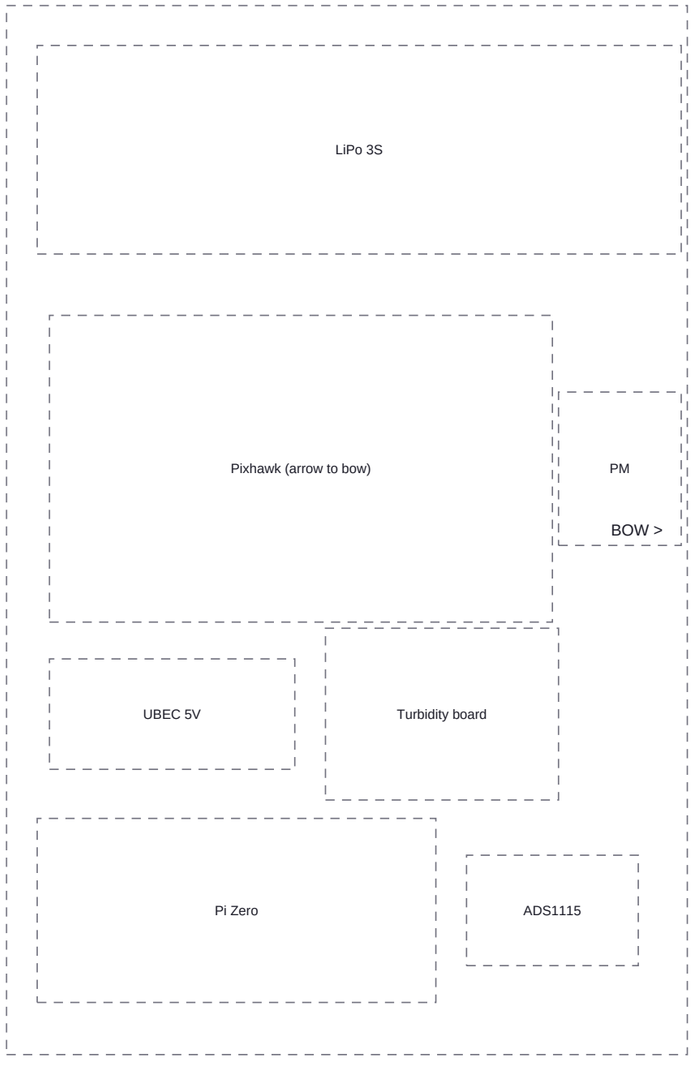
Bin bottom (drain holes)

3 mm sunboard | 110 x 220 mm



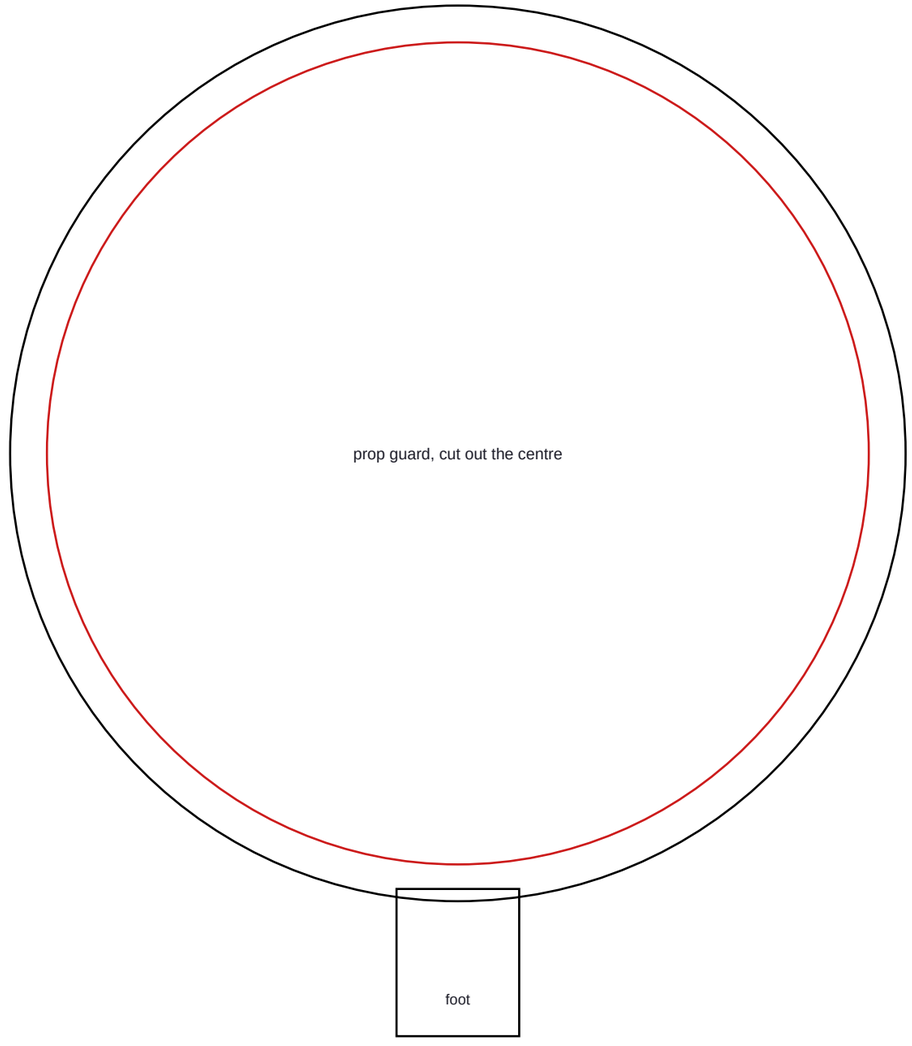
Electronics box floor guide

paper guide, glue inside the floor | 111 x 171 mm



Prop guard ring (left)

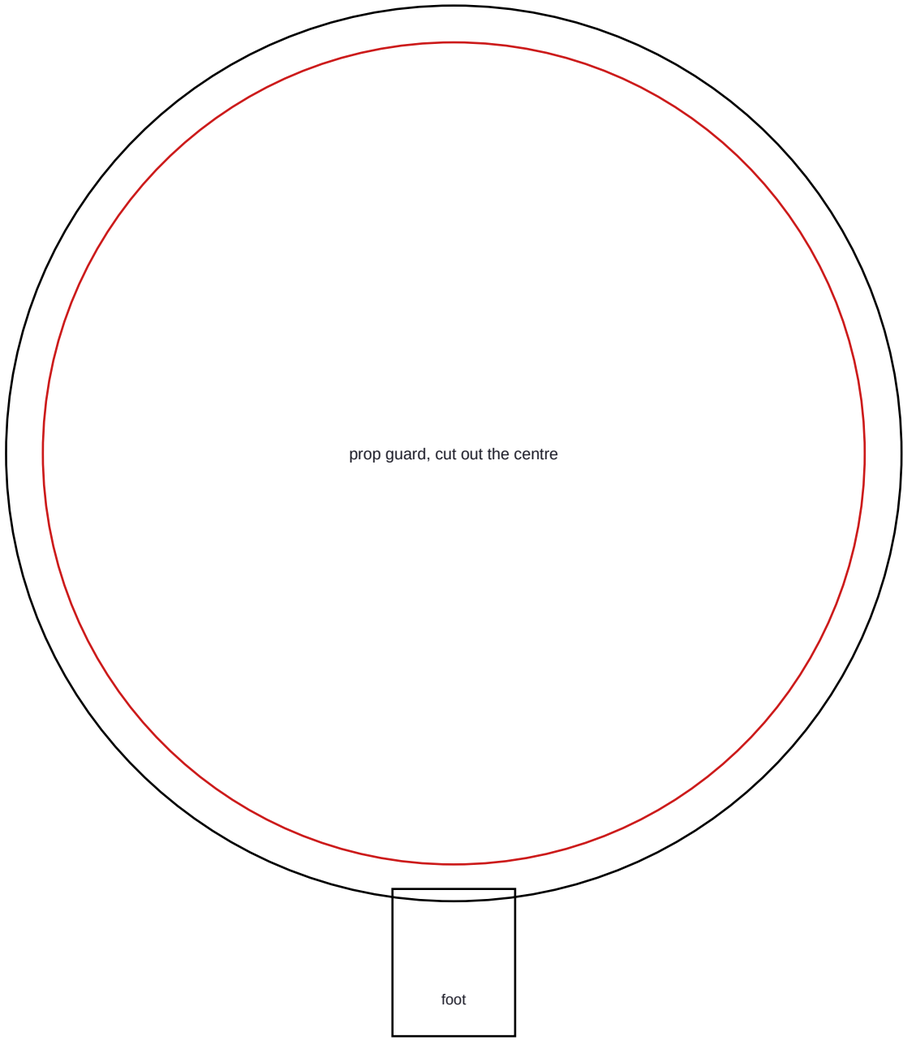
5 mm sunboard | 146 x 168 mm



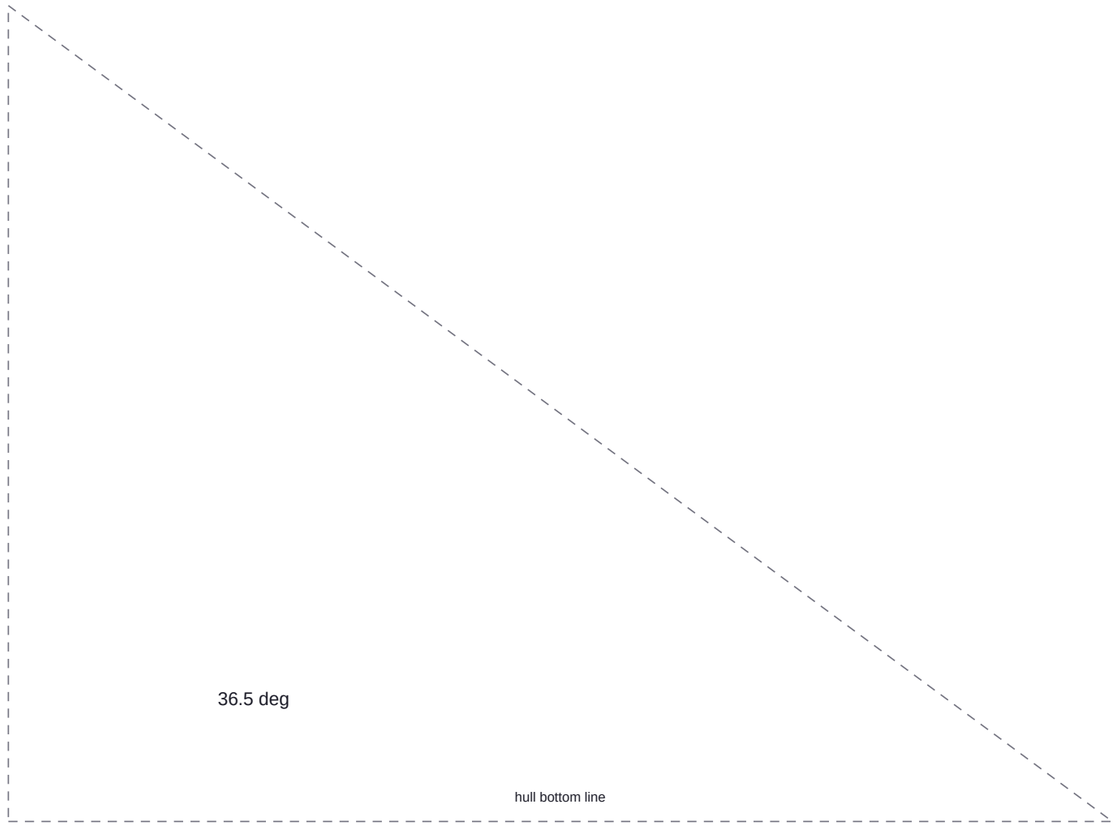
VARUNA Conveyor Cat | small parts 2 of 4

<- must measure 50 mm

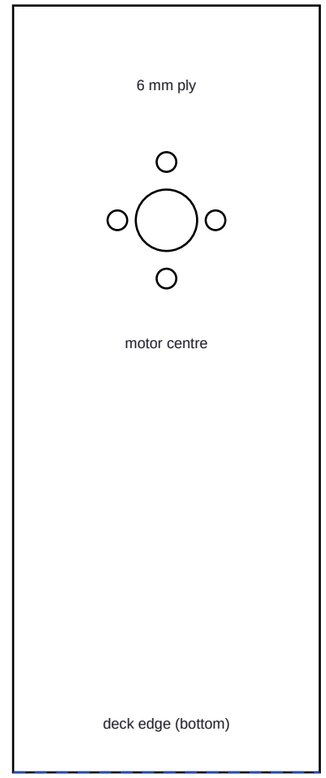
Prop guard ring (right)
5 mm sunboard | 146 x 168 mm



Conveyor angle guide
paper or card | 180 x 133 mm



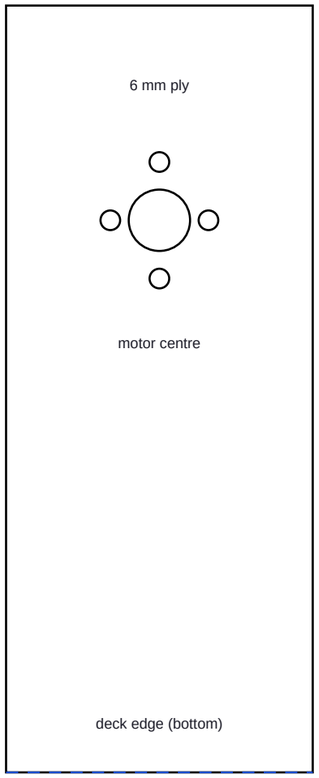
Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



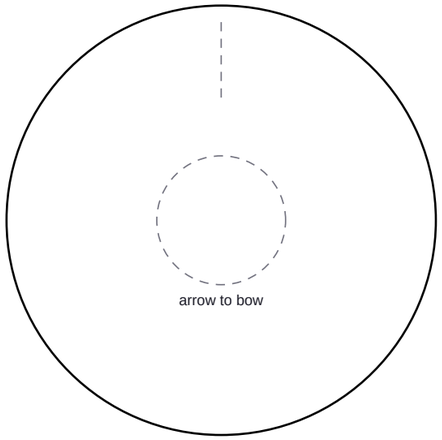
VARUNA Conveyor Cat | small parts 3 of 4

<- must measure 50 mm

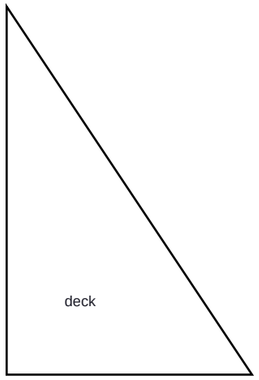
Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard) | 50 x 125 mm



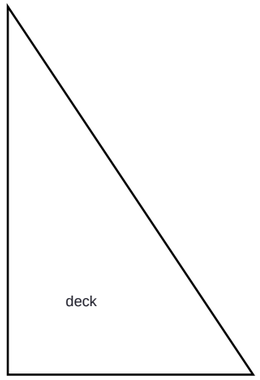
GPS disc
5 mm sunboard | 70 x 70 mm



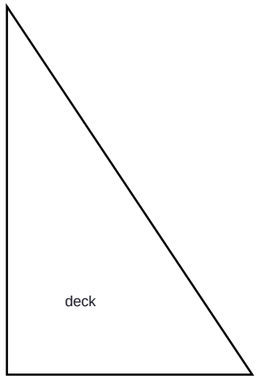
Tower gusset
5 mm sunboard | 40 x 60 mm



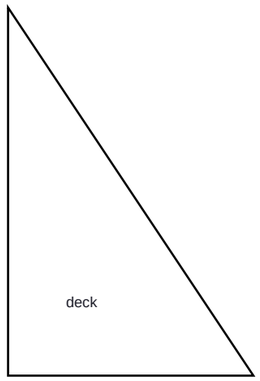
Tower gusset
5 mm sunboard | 40 x 60 mm



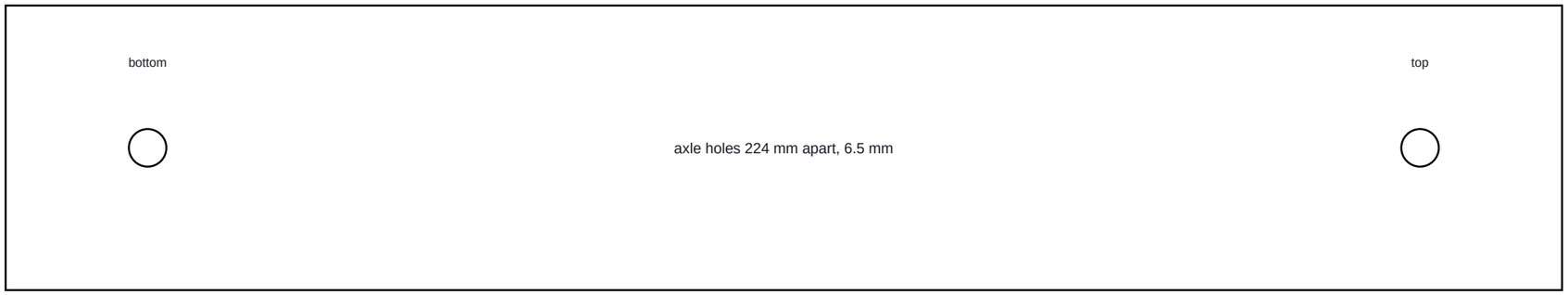
Tower gusset
5 mm sunboard | 40 x 60 mm



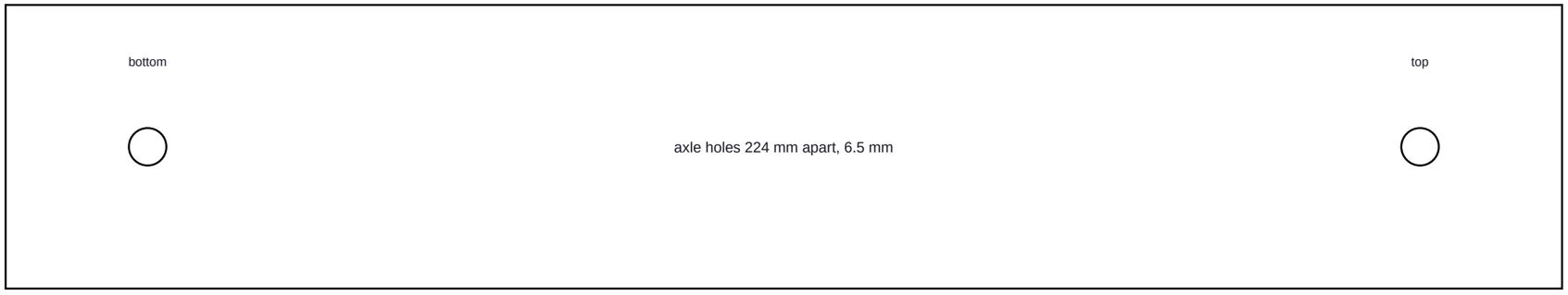
Tower gusset
5 mm sunboard | 40 x 60 mm



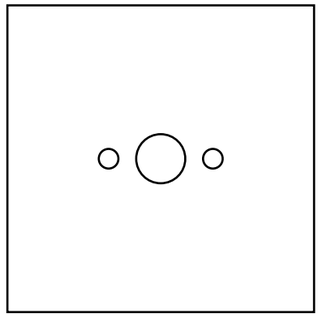
Conveyor side plate
5 mm sunboard (cut 2) | 274 x 50 mm



Conveyor side plate
5 mm sunboard (cut 2) | 274 x 50 mm



Gear motor bracket
5 mm sunboard | 50 x 50 mm



Bin wall front
3 mm sunboard | 220 x 45 mm

220 x 45

Bin wall back
3 mm sunboard | 220 x 45 mm

220 x 45

Bin wall side
3 mm sunboard | 104 x 42 mm

104 x 42

Bin wall side
3 mm sunboard | 104 x 42 mm

104 x 42

Mast foot ring
5 mm sunboard (cut 2, stack) | 40 x 40 mm

Mast foot ring
5 mm sunboard (cut 2, stack) | 40 x 40 mm

Camera plate
3 mm sunboard | 32 x 32 mm

lens up

Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm

35 deg

Camera wedge side (35 deg)
3 mm sunboard | 30 x 21 mm

35 deg

Roller end plug
5 mm sunboard (cut 8, stack 2 per end) | 21 x 21 mm

Roller end plug
5 mm sunboard (cut 8, stack 2 per end) | 21 x 21 mm

Roller end plug
5 mm sunboard (cut 8, stack 2 per end) | 21 x 21 mm

Roller end plug
5 mm sunboard (cut 8, stack 2 per end) | 21 x

Belt cleat
3 mm sunboard (cut 10) | 120 x 10 mm

Belt cleat
3 mm sunboard (cut 10) | 120 x 10 mm

Belt cleat
3 mm sunboard (cut 10) | 120 x 10 mm

Belt cleat
3 mm sunboard (cut 10) | 120 x 10 mm

Belt cleat
3 mm sunboard (cut 10) | 120 x 10 mm